SHORT-CIRCUIT CALCULATION WITH FULLSIZE CONVERTERS ACCORDING TO IEC 60909

Gerd Balzer
Darmstadt University of Technology
Landgraf-Georg-Str. 4, D-64283 Darmstadt, Germany, gerd.balzer@eev.tu-darmstadt.de

ABSTRACT

Due to the increasing integration of wind farms and PV systems in the electrical networks, it is necessary to consider the influence of these power plants on the short-circuit currents in transmission and distribution networks. In many cases, both the wind farm as well as the photovoltaic systems will be connected via VSC converters (full-size converters) with the power system. As the old IEC Standard (IEC 60909-0: 2001-07: Short-circuit currents in three-phase a.c. systems – Part 0: Calculation of currents) does not included the calculation of currents generated by self-commutated converters, a revision of this standard has been necessary. In the new version of IEC Standard Short-Circuit Calculation, the short-circuit current at the fault location is calculated in a first step, which is caused by the voltage sources, and the VSC converters are modeled with an infinite internal resistance in the positive sequence. In a second step, the existing voltage sources are shorted circuit, and the current contribution of the converters are determined to the short-circuit current at the fault location by using current sources. Subsequently, the superimposition of the two current components is carried out in order to determine the total current. Moreover, it is possible to determine the voltage distribution throughout the network. The maximum inductive current that the VSC converter can feed during the short duration must be provided by the manufacturer. All necessary currents can be determined based on the initial short-circuit alternating current, which are relevant for the design of equipment and installations: Peak short-circuit current, breaking short-circuit current, and steady state short-circuit current.

KEYWORDS: ***Short-circuit calculation, full-size converter, initial short-circuit current, renewal energy***

1 INTRODUCTION

According to the new IEC 60909-0 [1], the short-circuit current calculation can also be conducted in case of full-size converters, e.g. if the connection of wind and PV systems are considered. Figure 1 shows an example of the system circuit for these two cases. While the consideration of networks, generators and motors, uses voltage sources, a full-size converter has to be represented by a current source, which in some circumstances may feed in the (inductive) short-circuit current (voltage controlled). In this report it is shown, how the current calculation can be performed at the fault location in this case according to the new IEC standard, exemplarily in case of a three phase short circuit.

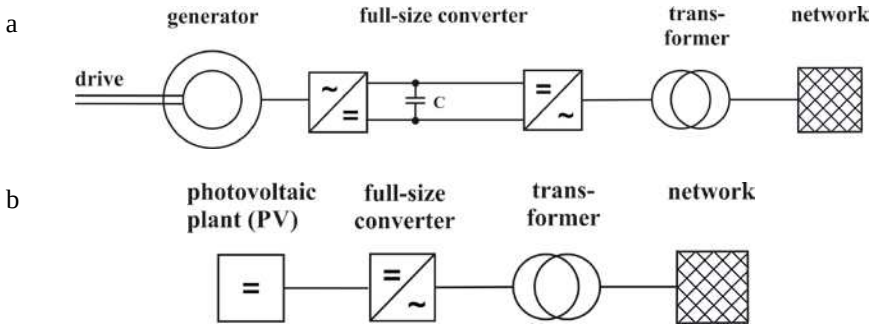


Figure 1: Installation of self-commutated converter
a: Wind power generation; b: Photo voltaic power generation

2 SIMULATION OF A POWER GENERATION WITH FULL-SIZE CONVERTERS

According to the new IEC standard, the full-size converter has to be simulated using the following circumstances:

- Power generation feeders with full-size converters (wind, photovoltaics) are modeled by a current source in the positive sequence. The current source depends on the short-circuit failure, and the value will be provided by the manufacturer. The shunt impedance of the power source $\underline{Z}_{PF}$ is assumed to be infinite.

- In case of unbalanced short circuits, the shunt impedance of the negative sequence system $\underline{Z}_{(2)PF}$ depends on the fault type as well as the control strategy. The zero sequence impedance $\underline{Z}_{(0)PF}$ can be considered as infinite.
- Power generation feeders can be neglected, if their contribution to the total initial short-circuit current without full-size converters is less than 5 %.

In the following sections, the equations for the three-phase short circuit are generally derived, in addition, the three-phase short circuit is calculated for a 110 kV system.

3 CALCULATION OF THE THREE PHASE IITIAL SHORT-CIRCUIT CURRENT

3.1 Procedure

As an example, the system circuit is used according to Fig. 2. In total, three wind parks are connected to the 110 kV network, which is supplied by the upstream system.

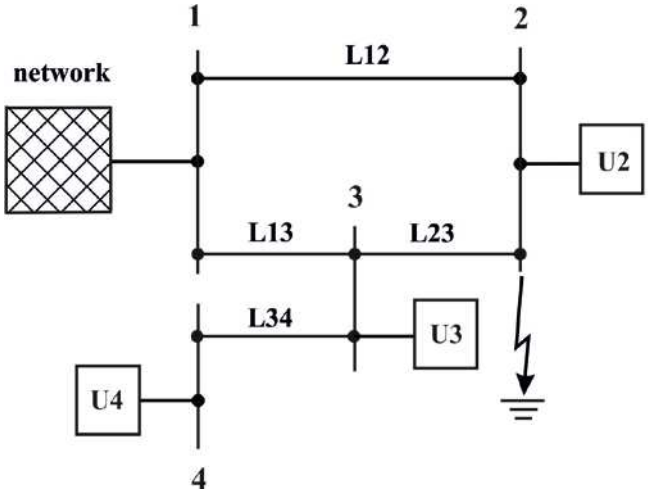


Figure 2: Equivalent circuit for the calculation of the short-circuit current

The following data of the components are assumed for the calculation of an example,

- Network
  - Initial short-circuit current: $I_{\mathrm{k}}^{''} = 20\,\mathrm{kA}$
  - *R*/*X* ratio: $R/X = 0{,}1$
- Overhead lines
  - impedance: $\underline{Z}_{\mathrm{L}}^{'} = (0{,}120 + \mathrm{j}0{,}393)\,\Omega/\mathrm{km}$
  - line lengths: $\ell_{12} = 100\,\mathrm{km}$, $\ell_{13} = \ell_{23} = 50\,\mathrm{km}$, $\ell_{34} = 25\,\mathrm{km}$
- Full-size converter (wind park)
  - rated power: $P_{\mathrm{U2}} = 100\,\mathrm{MW}$, $P_{\mathrm{U3}} = 50\,\mathrm{MW} = P_{\mathrm{U4}}$
  - short-circuit current: $I_{\mathrm{kU}} = k \cdot I_{\mathrm{rU}}$, $k = 1{,}2$

According to Fig. 2, the 110 kV system is assumed to be a voltage source, while the full-converters represent a constant current source. For this reason, in total two steps are used to calculate the short-circuit currents, namely:

- Calculation of short circuit current at the fault location $\underline{I}_{\mathrm{kI}i}^{''}$ (busbar 2), caused by the 110 kV system and
- calculation of short circuit current at the fault location $\underline{I}_{\mathrm{kII}i}^{''}$ (busbar 2), caused by the injection of the full –size converters.

The two parts of the short-circuit currents $\underline{I}_{\mathrm{kI}i}^{''}$ and $\underline{I}_{\mathrm{kII}i}^{''}$ are finally superimposed to determine the total short circuit current $\underline{I}_{\mathrm{k}i}^{''}$.

The equivalent circuit of the positive sequence of the three-phase short circuit is shown in Fig. 3, in this case both the current sources of full-size converter and the equivalent voltage source at the fault location F2 are depicted.

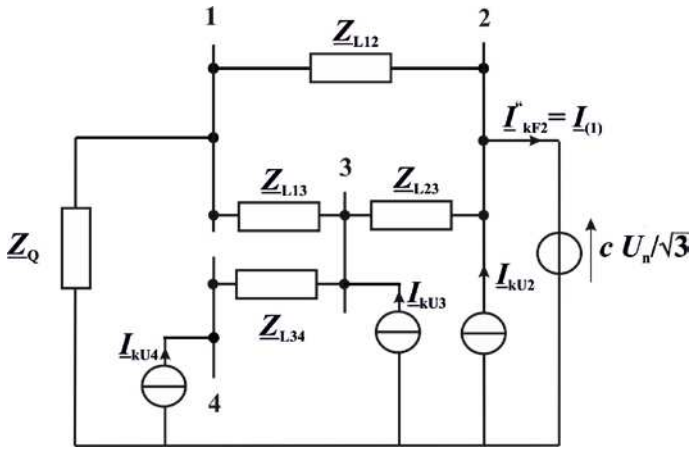


Figure 3: Equivalent circuit of the positive sequence system for the three phase short circuit at the location F2

3.2 Step 1: Equivalent voltage source at the fault location

Basically, the admittance matrix $\mathbf{Y}_N$ of the network is set up, followed by the calculation of the currents $\underline{\mathbf{I}}_i$ at the system node using the impedance matrix $\underline{\mathbf{Z}}_N$ and the voltage distribution $\underline{\mathbf{U}}_i$. In this connection, the current sources are considered to be infinite:

$$\underline{\mathbf{I}}_i = \underline{\mathbf{Y}}_N \cdot \underline{\mathbf{U}}_i \qquad\qquad \underline{\mathbf{U}}_i = \underline{\mathbf{Z}}_N \cdot \underline{\mathbf{I}}_i \tag{1}$$

Generally, the admittance resp. impedance matrix of a network becomes:

$$\underline{\mathbf{Y}}_N = \begin{pmatrix} \underline{Y}_{11} & \cdots & \underline{Y}_{1i} & \cdots & \underline{Y}_{1m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Y}_{i1} & \cdots & \underline{Y}_{ii} & \cdots & \underline{Y}_{im} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Y}_{m1} & \cdots & \underline{Y}_{mi} & \ddots & \underline{Y}_{mm} \end{pmatrix} \qquad \underline{\mathbf{Z}}_N = \begin{pmatrix} \underline{Z}_{11} & \cdots & \underline{Z}_{1i} & \cdots & \underline{Z}_{1m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{i1} & \cdots & \underline{Z}_{ii} & \cdots & \underline{Z}_{im} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{m1} & \cdots & \underline{Z}_{mi} & \ddots & \underline{Z}_{mm} \end{pmatrix} \tag{2}$$

Fig. 4 applies to the equivalent voltage source at the fault location, with the amplitude of the equivalent voltage source according to equation (3).

$$\underline{U}_i = c \cdot U_n / \sqrt{3} \tag{3}$$

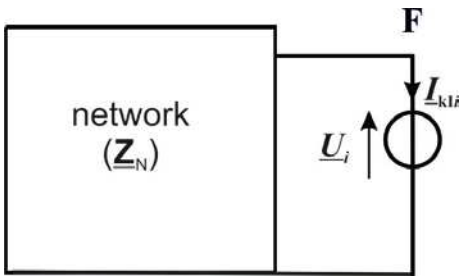


Figure 4: Equivalent circuit with the equivalent voltage source at the fault location

Using the impedance matrix, it is possible to calculate the short-circuit current at the fault location. Here, the equivalent voltage is applied at node *i* according to equation (3), while all currents $\underline{I}_i$ of the system nodes are equal to zero, except for the fault current. Generally, the equation becomes:

$$\begin{pmatrix} \underline{U}_1 \\ \vdots \\ \underline{U}_i \\ \vdots \\ \underline{U}_m \end{pmatrix} = \begin{pmatrix} \underline{Z}_{11} & \cdots & \underline{Z}_{1i} & \cdots & \underline{Z}_{1m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{i1} & \cdots & \underline{Z}_{ii} & \cdots & \underline{Z}_{im} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{m1} & \cdots & \underline{Z}_{mi} & \ddots & \underline{Z}_{mm} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ \vdots \\ \underline{I}''_{kIi} \\ \vdots \\ 0 \end{pmatrix} \tag{4}$$

From equation (4), the short-circuit current $\underline{I}''_{\mathrm{k}Ii}$ results without the influence of full-size converters to:

$$c \cdot U_\mathrm{n} / \sqrt{3} = \underline{Z}_{ii} \cdot \underline{I}''_{\mathrm{k}Ii} \qquad \underline{I}''_{\mathrm{k}Ii} = \frac{c \cdot U_\mathrm{n} / \sqrt{3}}{\underline{Z}_{ii}} \tag{5}$$

Subsequently, all node voltages related to ground can be derived from equations (4), so that, for example, the voltage $\underline{U}_1$ at the system node 1 applies in case of a short circuit at node *i*:

$$\underline{U}_1 = \underline{Z}_{1i} \cdot \underline{I}''_{\mathrm{k}Ii} = \frac{\underline{Z}_{1i}}{\underline{Z}_{ii}} \cdot c \cdot \underline{U}_\mathrm{n} / \sqrt{3} \tag{6}$$

It should be noted that the voltage according to equation (6) is obtained as a result of the changed voltage profile due to the equivalent voltage source at the fault location. The actual voltage $\underline{U}_{1\mathrm{r}}$ becomes:

$$\underline{U}_{1\mathrm{r}} = c \cdot \underline{U}_\mathrm{n} / \sqrt{3} - \frac{\underline{Z}_{1i}}{\underline{Z}_{ii}} \cdot c \cdot \underline{U}_\mathrm{n} / \sqrt{3} = c \cdot \underline{U}_\mathrm{n} / \sqrt{3} \cdot \left(1 - \frac{\underline{Z}_{1i}}{\underline{Z}_{ii}} \right) \tag{7}$$

3.3 Step 2: Constant current source

The representation of the full-size converters is performed by a current source with the internal impedance $\underline{Z}_{\mathrm{U}i} \rightarrow \infty$, so that the equivalent circuit is based on Figure 5.

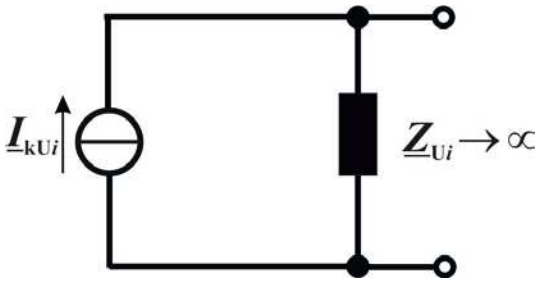


Figure 5: Representation of self-commutated converter as current source (positive sequence system)

In a similar way, according to the method of the equivalent voltage source at the fault location (equation 4), the current distribution is calculated exclusively by the influence of full-size converters, so that the system of equations is considered according to equation (8). At the fault location, the voltage $\underline{U}_\mathrm{i}$ is short circuited, while the remaining voltages at the system nodes are calculated due to the current conditions.

$$\begin{pmatrix} \underline{U}_1 \\ \vdots \\ 0 \\ \vdots \\ \underline{U}_\mathrm{m} \end{pmatrix} = \begin{pmatrix} \underline{Z}_{11} & \cdots & \underline{Z}_{1i} & \cdots & \underline{Z}_{1\mathrm{m}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{i1} & \cdots & \underline{Z}_{ii} & \cdots & \underline{Z}_{i\mathrm{m}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{\mathrm{m}1} & \cdots & \underline{Z}_{\mathrm{m}i} & \ddots & \underline{Z}_{\mathrm{mm}} \end{pmatrix} \cdot \left[\begin{pmatrix} 0 \\ \vdots \\ \underline{I}''_{\mathrm{kII}i} \\ \vdots \\ 0 \end{pmatrix} - \begin{pmatrix} \underline{I}_{\mathrm{kU}1} \\ \vdots \\ \underline{I}_{\mathrm{kU}i} \\ \vdots \\ \underline{I}_{\mathrm{kU}\mathrm{m}} \end{pmatrix} \right] \tag{8}$$

As a result of the power feed-in of the full-size converters, the short-circuit contribution $\underline{I}''_{\mathrm{kII}i}$ at the fault location becomes:

$$0 = \underline{Z}_{ii} \cdot \underline{I}''_{\mathrm{kII}i} - \left(\underline{Z}_{i1} \cdot \underline{I}_{\mathrm{U}1} + \cdots + \underline{Z}_{i\mathrm{m}} \cdot \underline{I}_{\mathrm{kU}\mathrm{m}} \right) = \underline{Z}_{ii} \cdot \underline{I}''_{\mathrm{kII}i} - \sum_j^\mathrm{m} \underline{Z}_{\mathrm{i}\mathrm{j}} \cdot \underline{I}_{\mathrm{kU}j} \tag{9}$$

After conversion:

$$\underline{I}''_{\mathrm{kII}i} = \frac{1}{\underline{Z}_{ii}} \cdot \sum_j^\mathrm{m} \underline{Z}_{ij} \cdot \underline{I}_{\mathrm{kU}j} \tag{10}$$

3.4 Step 3: Superposition

The two current components according to the equations (5) and (10) are added together to calculate the total current at the fault location, so that the following equation (11) applies.

$$\underline{I}''_{\mathrm{k}i} = \underline{I}''_{\mathrm{kI}i} + \underline{I}''_{\mathrm{kII}i} = \frac{c \cdot U_{\mathrm{n}} / \sqrt{3}}{\underline{Z}_{ii}} + \frac{1}{\underline{Z}_{ii}} \cdot \sum_{j}^{\mathrm{m}} \underline{Z}_{ij} \cdot \underline{I}_{\mathrm{kU}j} \tag{11}$$

The values $\underline{Z}_{ij}$ $\underline{Z}_{ii}$ are derived from the impedance matrix according to equation (2), where *i* represents the short-circuit location and *j* the node, which the full-size converter is connected to the system. Complex values are used to determine the total current, whereas the amounts of the currents are considered according to the IEC Standard [1].

3.5 Step 4: Calculation of the voltage distribution

Starting from the current distribution, the voltages can be determined at the network nodes. Here, the system of equations (12) is used taking the entire short-circuit current $\underline{I}''_{\mathrm{k}i}$ according section 3.3 into consideration.

$$\begin{pmatrix} \underline{U}_1 \\ \vdots \\ \underline{U}_i \\ \vdots \\ \underline{U}_{\mathrm{m}} \end{pmatrix} = \begin{pmatrix} \underline{Z}_{11} & \cdots & \underline{Z}_{1i} & \cdots & \underline{Z}_{1\mathrm{m}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{i1} & \cdots & \underline{Z}_{ii} & \cdots & \underline{Z}_{i\mathrm{m}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \underline{Z}_{\mathrm{m}1} & \cdots & \underline{Z}_{\mathrm{m}i} & \ddots & \underline{Z}_{\mathrm{mm}} \end{pmatrix} \cdot \left[\begin{pmatrix} 0 \\ \vdots \\ \underline{I}''_{\mathrm{k}i} \\ \vdots \\ 0 \end{pmatrix} - \begin{pmatrix} \underline{I}_{\mathrm{kU}1} \\ \vdots \\ \underline{I}_{\mathrm{kU}i} \\ \vdots \\ \underline{I}_{\mathrm{kU}\mathrm{m}} \end{pmatrix} \right] \tag{12}$$

$\underline{U}_{\mathrm{i}}$ is the equivalent voltage source at the fault location according to equation (3). Even in this case, the calculated voltages represent the voltage changes due to the short circuit. The real voltages $\underline{U}_{i\mathrm{r}}$ at the system nodes are calculated according equation (7).

3.6 Example

Based on the impedances of the equipment specified in section 3.1, the corresponding matrices are set up to determine the initial short-circuit current according to equation 11. The following impedances can be calculated due to the rated values (section 3.1):

110 kV system:

$$Z_{\mathrm{Q}} = \frac{c \cdot U_{\mathrm{nQ}}}{\sqrt{3} \cdot I''_{\mathrm{k}}} = \frac{1{,}1 \cdot 110}{\sqrt{3} \cdot 20} \cdot \Omega = 3{,}4930\,\Omega \qquad \underline{Z}_{\mathrm{Q}} = (0{,}3476 + \mathrm{j}3{,}4756)\Omega$$

Overhead lines:

$$\underline{Z}_{\mathrm{L12}} = (12{,}000 + \mathrm{j}39{,}300)\Omega \qquad \underline{Z}_{\mathrm{L13}} = (6{,}000 + \mathrm{j}19{,}650)\Omega = \underline{Z}_{\mathrm{L12}} / 2$$

$$\underline{Z}_{\mathrm{L23}} = (6{,}000 + \mathrm{j}19{,}650)\Omega = \underline{Z}_{\mathrm{L12}} / 2 \qquad \underline{Z}_{\mathrm{L34}} = (3{,}000 + \mathrm{j}9{,}825)\Omega = \underline{Z}_{\mathrm{L12}} / 4$$

Full-size converters (wind parks)

$$I_{\mathrm{rU2}} = \frac{P_{\mathrm{U2}}}{\sqrt{3} \cdot U_{\mathrm{n}}} = \frac{100}{\sqrt{3} \cdot 110}\,\mathrm{kA} = 0{,}5249\,\mathrm{kA} \qquad I_{\mathrm{rU3}} = 0{,}2624\,\mathrm{kA} = I_{\mathrm{rU4}}$$

As the injected current of the full-size converts is assumed to be inductive, the respective currents are arising in case oof a short circuit to:

$$\underline{I}_{\mathrm{kU2}} = -\mathrm{j}0{,}6298\,\mathrm{kA} \qquad \underline{I}_{\mathrm{kU3}} = -\mathrm{j}0{,}3149\,\mathrm{kA} = \underline{I}_{\mathrm{kU4}}$$

The admittance matrix can be derived according equation (2) for the example of Fig. 2.

$$\underline{\mathbf{Y}}_{\mathrm{N}} = \begin{pmatrix} \underline{Y}_{\mathrm{Q}} + \underline{Y}_{\mathrm{L12}} + \underline{Y}_{\mathrm{L13}} & -\underline{Y}_{\mathrm{L12}} & -\underline{Y}_{\mathrm{L13}} & 0 \\ -\underline{Y}_{\mathrm{L12}} & \underline{Y}_{\mathrm{L12}} + \underline{Y}_{\mathrm{L23}} & -\underline{Y}_{\mathrm{L23}} & 0 \\ -\underline{Y}_{\mathrm{L13}} & -\underline{Y}_{\mathrm{L23}} & \underline{Y}_{\mathrm{L13}} + \underline{Y}_{\mathrm{L23}} + \underline{Y}_{\mathrm{L34}} & -\underline{Y}_{\mathrm{L34}} \\ 0 & 0 & -\underline{Y}_{\mathrm{L34}} & \underline{Y}_{\mathrm{L34}} \end{pmatrix} \tag{13}$$

Due to the considered values, the equation becomes:

$$\underline{\mathbf{Y}}_{\mathrm{N}} = \begin{pmatrix} +0{,}04981 - \mathrm{j}0{,}35470 & -0{,}00711 + \mathrm{j}0{,}02328 & -0{,}01421 + \mathrm{j}0{,}04655 & 0 \\ -0{,}00711 + \mathrm{j}0{,}02328 & +0{,}02132 - \mathrm{j}0{,}06983 & -0{,}01421 + \mathrm{j}0{,}04655 & 0 \\ -0{,}01421 + \mathrm{j}0{,}04655 & -0{,}01421 + \mathrm{j}0{,}04655 & +0{,}05685 - \mathrm{j}0{,}18620 & -0{,}02843 + \mathrm{j}0{,}09310 \\ 0 & 0 & -0{,}02843 + \mathrm{j}0{,}09310 & +0{,}02843 - \mathrm{j}0{,}09310 \end{pmatrix}$$

The inversion of the admittance matrix leads to the impedance matrix to calculate the short-circuit currents.

$$\underline{Z}_{\mathrm{N}} = \begin{pmatrix} +0{,}3476 + \mathrm{j}3.4756 & +0{,}3476 + \mathrm{j}3.4756 & +0{,}3476 + \mathrm{j}3.4756 & +0{,}3476 + \mathrm{j}3.4756 \\ +0{,}3476 + \mathrm{j}3.4756 & +6{,}3469 + \mathrm{j}23{,}1238 & +3{,}3473 + \mathrm{j}13{,}2997 & +3{,}3473 + \mathrm{j}13{,}2997 \\ +0{,}3476 + \mathrm{j}3.4756 & +3{,}3473 + \mathrm{j}13{,}2997 & +4{,}8468 + \mathrm{j}18{,}2132 & +4{,}8468 + \mathrm{j}18{,}2132 \\ +0{,}3476 + \mathrm{j}3.4756 & +3{,}3473 + \mathrm{j}13{,}2997 & +4{,}8468 + \mathrm{j}18{,}2132 & +7{,}8471 + \mathrm{j}28{,}0381 \end{pmatrix}$$

The calculation on the basis of the example is carried out with complex quantities in accordance with equation (11), in contrast to the IEC standard. The total short-circuit current at the location F2 (i = 2, busbar 2) becomes:

$$\underline{I}''_{\mathrm{kF2}} = \frac{c \cdot U_{\mathrm{n}} / \sqrt{3}}{\underline{Z}_{22}} + \frac{1}{\underline{Z}_{22}} \cdot \left[\underline{Z}_{21} \cdot \underline{I}_{\mathrm{kU1}} + \underline{Z}_{22} \cdot \underline{I}_{\mathrm{kU2}} + \underline{Z}_{23} \cdot \underline{I}_{\mathrm{kU3}} + \underline{Z}_{24} \cdot \underline{I}_{\mathrm{kU4}} \right] \tag{14}$$

As no full-size converter feeds in at busbar (1) ($\underline{I}_{\mathrm{kU1}} = 0$) , the short-circuit current is calculated by simplification:

$$\underline{I}''_{\mathrm{kF2}} = \frac{c \cdot U_{\mathrm{n}} / \sqrt{3}}{\underline{Z}_{22}} + \frac{1}{\underline{Z}_{22}} \cdot \left[\underline{Z}_{23} \cdot \underline{I}_{\mathrm{kU3}} + \underline{Z}_{24} \cdot \underline{I}_{\mathrm{kU4}} \right] + \underline{I}_{\mathrm{kU2}} \tag{15}$$

After insertion of the numerical values, equation (12) becomes with $\underline{Z}_{23} = Z_{24}$ and $\underline{I}_{\mathrm{kU3}} = \underline{I}_{\mathrm{kU4}}$:

$$\underline{I}''_{\mathrm{kF2}} = \frac{1{,}1 \cdot 110 / \sqrt{3}}{6{,}3469 + \mathrm{j}23{,}1238}\,\mathrm{kA} - \mathrm{j}\frac{3{,}3473 + \mathrm{j}13{,}2997}{6{,}3469 + \mathrm{j}23{,}1238} \cdot 0{,}6298\,\mathrm{kA} - \mathrm{j}0{,}6298\,\mathrm{kA} = 0{,}7788\,\mathrm{kA} - \mathrm{j}3{,}7994\,\mathrm{kA}$$

The first part relates to the contribution, which is caused by the 110 kV network ($I''_{\mathrm{kIF2}} = 2{,}9133\,\mathrm{kA}$), while the second and third part is related to the feed-in of the full-size converter ($I''_{\mathrm{kIIF2}} = 0{,}9899\,\mathrm{kA}$). The amount of the total current is:

$$I''_{\mathrm{kF2}} = 3{,}8782\,\mathrm{kA}$$

If the current is calculated according to the IEC Standard with the amounts of the currents, the result becomes:

$$I''_{\mathrm{kF2}} = \frac{1{,}1 \cdot 110 / \sqrt{3}}{23{,}9790}\,\mathrm{kA} + \frac{13{,}7144}{23{,}9790} \cdot 0{,}6298\,\mathrm{kA} + 0{,}6298\,\mathrm{kA} = 2{,}9134\,\mathrm{kA} + 0{,}9900\,\mathrm{kA} = 3{,}9034\,\mathrm{kA}$$

In this case, the calculation with amounts provides a result which is greater by 0.65% as the value with complex values. The voltage distribution at the network nodes may be determined with the aid of the equation system (16).

$$\begin{pmatrix} \underline{U}_1 \\ \underline{U}_2 \\ \underline{U}_3 \\ \underline{U}_4 \end{pmatrix} = \begin{pmatrix} \underline{Z}_{11} & \underline{Z}_{12} & \underline{Z}_{13} & \underline{Z}_{14} \\ \underline{Z}_{21} & \underline{Z}_{22} & \underline{Z}_{23} & \underline{Z}_{24} \\ \underline{Z}_{31} & \underline{Z}_{32} & \underline{Z}_{33} & \underline{Z}_{34} \\ \underline{Z}_{41} & \underline{Z}_{42} & \underline{Z}_{43} & \underline{Z}_{44} \end{pmatrix} \cdot \left[\begin{pmatrix} 0 \\ I''_{\mathrm{kF2}} \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ \underline{I}_{\mathrm{kU2}} \\ \underline{I}_{\mathrm{kU3}} \\ \underline{I}_{\mathrm{kU4}} \end{pmatrix} \right] \tag{16}$$

The single voltage nodes can be considered taking into account the short-circuit current according to equation (15):

$$\begin{aligned} \underline{U}_1 &= \frac{\underline{Z}_{12}}{\underline{Z}_{22}} \cdot \left[c \cdot U_{\mathrm{n}} / \sqrt{3} + 2 \cdot \left(\underline{Z}_{23} - \underline{Z}_{22} \right) \cdot \underline{I}_{\mathrm{kU3}} \right] \\ \underline{U}_2 &= c \cdot U_{\mathrm{n}} / \sqrt{3} \\ \underline{U}_3 &= \frac{\underline{Z}_{23}}{\underline{Z}_{22}} \cdot \left[c \cdot U_{\mathrm{n}} / \sqrt{3} + 2 \cdot \left(\underline{Z}_{23} - \frac{\underline{Z}_{22} \cdot \underline{Z}_{33}}{\underline{Z}_{23}} \right) \cdot \underline{I}_{\mathrm{kU3}} \right] \\ \underline{U}_4 &= \frac{\underline{Z}_{23}}{\underline{Z}_{22}} \cdot \left[c \cdot U_{\mathrm{n}} / \sqrt{3} + \underline{I}_{\mathrm{kU3}} \cdot \left(2 \cdot \underline{Z}_{23} - \frac{\underline{Z}_{22} \cdot \left(\underline{Z}_{33} + \underline{Z}_{44} \right)}{\underline{Z}_{23}} \right) \right] \end{aligned} \tag{17}$$

The voltages with full-size converters are calculated taking into account the values of the impedance matrix, in this case the real distribution of the voltages U_{ir} is considered according to equation (7):

$$U_{1\mathrm{r}} = 0{,}8702 \cdot c \cdot U_{\mathrm{n}} / \sqrt{3} \qquad\qquad U_{2\mathrm{r}} = 0{,}0$$

$U_{3r} = 0{,}5250 \cdot c \cdot U_n / \sqrt{3}$ $U_{4r} = 0{,}5703 \cdot c \cdot U_n / \sqrt{3}$

If the full-size converters are not included, the node voltages become:

$U_{1r} = 0{,}8567 \cdot c \cdot U_n / \sqrt{3}$ $U_{2r} = 0{,}0$

$U_{3r} = U_{4r} = 0{,}4284 \cdot c \cdot U_n / \sqrt{3}$

The node voltages arise by inductive feed-in of the full-size converter during the short-circuit phase, this is true of course not for the short-circuit location.

4 PEAK SHORT-CIRCUIT CURRENT

4.1 Calculation of the peak short-circuit current

In each case, the peak short-circuit current is superposed of two parts (without and with full-size converter) to get the total current. The peak short-circuit current of a full-size converter is calculated according to equation (18) to:

$$i_p = \sqrt{2} \cdot I''_{kU} \quad (18)$$

With

I''_{kU} (= I_{sU}) short-circuit current contribution of a full-size converter on the high voltage side, value will be provided by the manufacturer, equation (10).

In a radial network, the resulting peak short-circuit current is calculated by the sum the partial currents. Assuming a mesh network, in contrast to equation (18), the resulting current is obtained at any network node by equation (19):

$$i_p = \kappa \cdot \sqrt{2} \cdot I''_{kI} + \sqrt{2} \cdot I''_{kII} \quad (17)$$

With

I''_{kI} three phase initial short-circuit current at the fault location without contribution of the full-size converter according equation (5).

I''_{kII} contribution of the full-size converter to the three phase short-circuit current at the fault location, equation (10).

When calculating the peak short-circuit current according to equation (19), a DC component of the current without the full-size converter is taken into considerations (κ factor), but not when determining the contribution of the full-size converter. The example of Figure 6 shows the transient short-circuit current curve in case of a short circuit in the 380 kV network. Here, a three-phase fault at the end of a short transmission line is assumed, which is initiated at the zero crossing of voltage phase R.

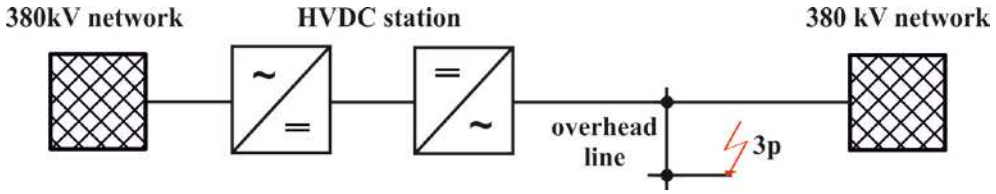


Figure 6: Equivalent circuit for the calculation of the three phase short-circuit current of a HVDC installation

Figure 7 shows the current transients of conductor R of the network feeder (Fig. 7a) and the HVDC part (Fig. 7b). In addition, the RMS value is listed, which is calculated from the three phase voltages.

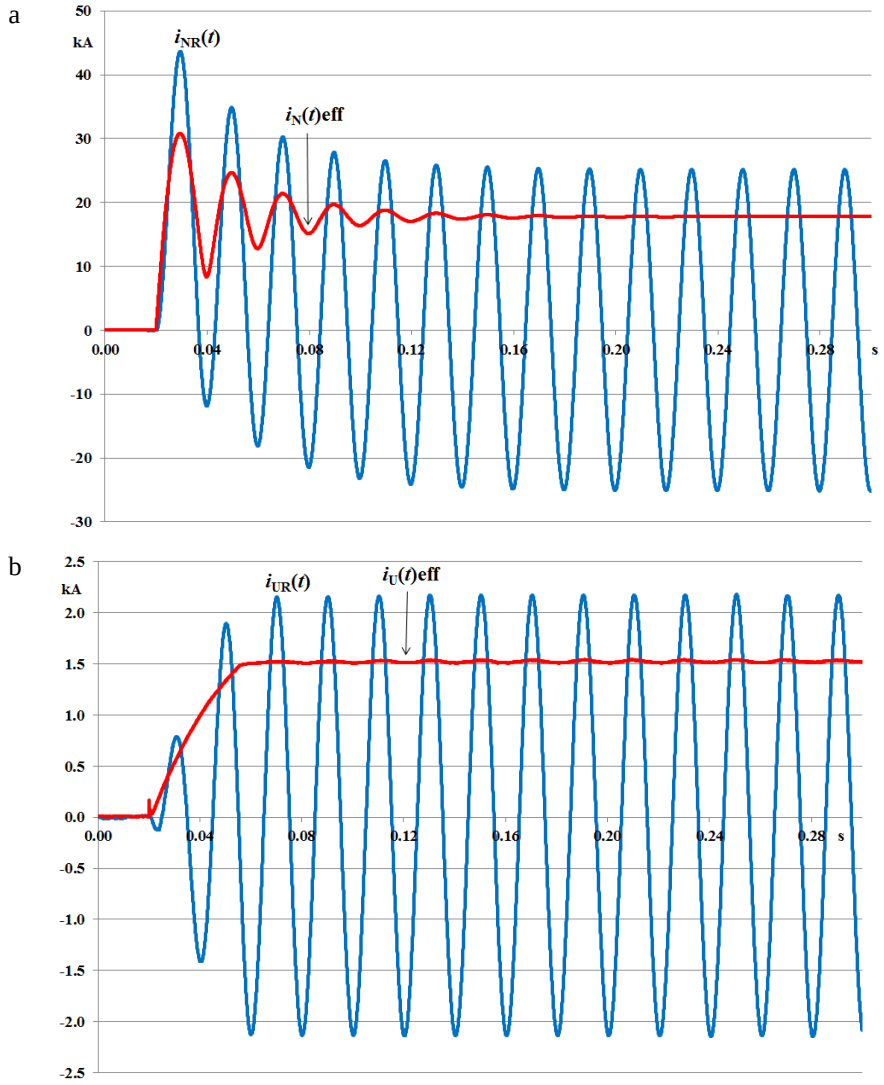


Figure 7:Current transient of phase R in case of a three phase short circuit
a: System contribution (N); b: HVDC contribution (U)

After short circuit initiation, the current contribution of the HVDC network is controlled so that the rated current is fed-in as an inductive current. Figure 8 shows the superposition of the current transients in each phase R. This inductive current is superimposed to the network part at the fault location, which is predominantly inductive current, due to the *R*/*X* ratio.

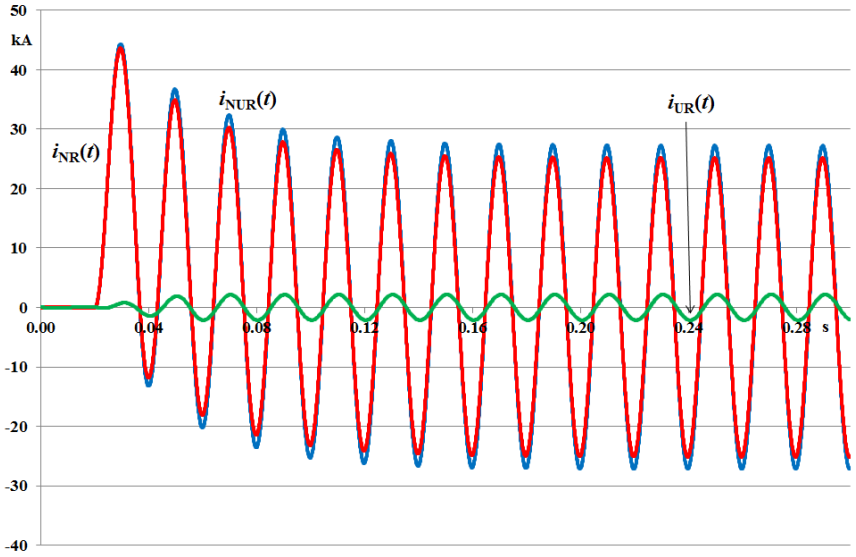


Figure 8: Superposition of the transients phase R
$i_{NR}(t)$: Network; $i_{UR}(t)$:HVDC; $i_{NUR}(t)$: total current

The following results can be obtained from Figures 7 and 8:

- The fault current is primarily determined by the network part, which results from the ratio of the currents I_k'' / I_{rU}.
- The part of the full-size converter has no DC component, so that equation (19) is justified.

In this example, the maximum value of the fault current is increased from i_p = 43,622 kA to i_p = 44,324 kA, which represents an increase of 1,609%.

4.2 Calculation of the peak short-circuit current (example)
The peak short-circuit current is calculated according to equation (19) considering the two parts of the initial short-circuit current. In case of a short circuit at node 2 of Fig. 2, the system is a radial network (the parallel circuit consists of impedances with the same *R*/*X* ratio), and in consequence, the factor κ can be obtained from the short-circuit impedance at the fault location. The resulting short-circuit impedance $\underline{Z}_{22}$ is derived from the impedance matrix.

$$\underline{Z}_{22} = (6{,}3469 + \mathrm{j}23.1238)\Omega \qquad R/X = 0{,}2745 \qquad \kappa = 1{,}4501$$

The peak short-circuit current becomes:

$$i_p = 1{,}4501 \cdot \sqrt{2} \cdot 2{,}9133\,\mathrm{kA} + \sqrt{2} \cdot 0{,}9899\,\mathrm{kA} = 5{,}9746\,\mathrm{kA} + 1{,}3999\,\mathrm{kA} = 7{,}3746\,\mathrm{kA}$$

5 BREAKING AND STEADY STATE CURRENT
The three phase breaking current as well as the steady state current are equal to the initial short-circuit current.

6 CONCLUSION
With the new IEC Standard, it is possible to calculate the short-circuit contribution of VSC converters (full-size converters), which connect the wind and PV systems to the distribution and transmission grid. In this report, only three-phase short-circuit current are presented, but the new IEC standard covers all types of short circuits.

The next step is to take the contribution of HVDC systems to the network short-circuit current into account using a similar calculation pattern.

REFERENCES

[1] IEC 60909-0:2016 Edition 2.0 (2016-01-28): Short-circuit current calculations in three-phase a.c. systems – Part 0: Calculation of currents